

Element G Report

Construction of a Testable Prototype

Problem Statement

Reports indicate that 42,000 American homeowners annually face recurring neck, shoulder, and back injuries from picking up and disposing of leaves in their yards¹. Disregarding this task causes leaves to accumulate in yards, smothering the grass and blocking the sunlight needed for proper growth. Overall, leaf removal is a task that can cause physical strain when done and, if neglected, can affect the yard's health, leading to grass death in the fall.

Bill of Materials

The bill of materials (BOM) lists every component required to build the lawn sweeper, including the quantity, vendor, and relevant notes for each item. Based on the research conducted, the listed parts are necessary to construct the lawn sweeper capable of efficiently collecting leaves. After accounting for materials already available and shipping costs, the total estimated cost is \$92.71.

Item	Qty	Description	Seller	Cost	Notes
Axle	3	8mm REX® Shaft with E-Clip (Stainless Steel, 624mm Length)	goBILDA	\$53.97	
Bearing	3	1611 Series Flanged Ball Bearing-8mm REX® ID x 14mm OD, 5mm Thickness	goBILDA	\$0.00	Group member already has this product
Belts	1	2mm Pitch GT2 Timing Belt (6mm Width, 360mm Pitch Length, 180 Tooth)	goBILDA	\$6.29	
Pulley	1	2mm Pitch GT2 Pinion Timing Pulley (8mm REX® Bore, 40 Tooth)	goBILDA	\$9.99	Big Pulley
Pulley	1	2mm Pitch GT2 Pinion Timing Pulley (8mm REX® Bore, 20 Tooth)	goBILDA	\$7.99	Small Pulley
Hood	1	3D Printed Hood	NVHS 3D Printers	\$0.00	Will be 3D Printed
Wheel	2	3D Printed Wheels	NVHS 3D Printers	\$0.00	Will be 3D Printed
Body	1	3D Printed Body	NVHS 3D Printers	\$0.00	Will be 3D Printed
Bag Frame	1	3D Printed Frame	NVHS 3D Printers	\$0.00	Will be 3D Printed
Intake Spinner	1	3D Printed	NVHS 3D Printers	\$0.00	Will be 3D Printed
Spinner Claws	28	3D Printed Claws for Intake	NVHS 3D Printers	\$0.00	Will be 3D Printed
Handle	1	3D Printed Grip Handle	NVHS 3D Printers	\$0.00	Will be 3D Printed
Shipping Costs				11.99	
Total Cost				\$92.71	

¹ Don't let raking leaves become a pain in the neck, back or spine. Luminis Health. (n.d.).
<https://www.luminishealth.org/en/blog/dont-let-raking-leaves-become-a-pain-in-the-neck-back-or-spine?>

Below is an exploded view drawing (Figure G.1) of the lawn sweeper, and labeled major components in the parts list table underneath (Figure G.2)

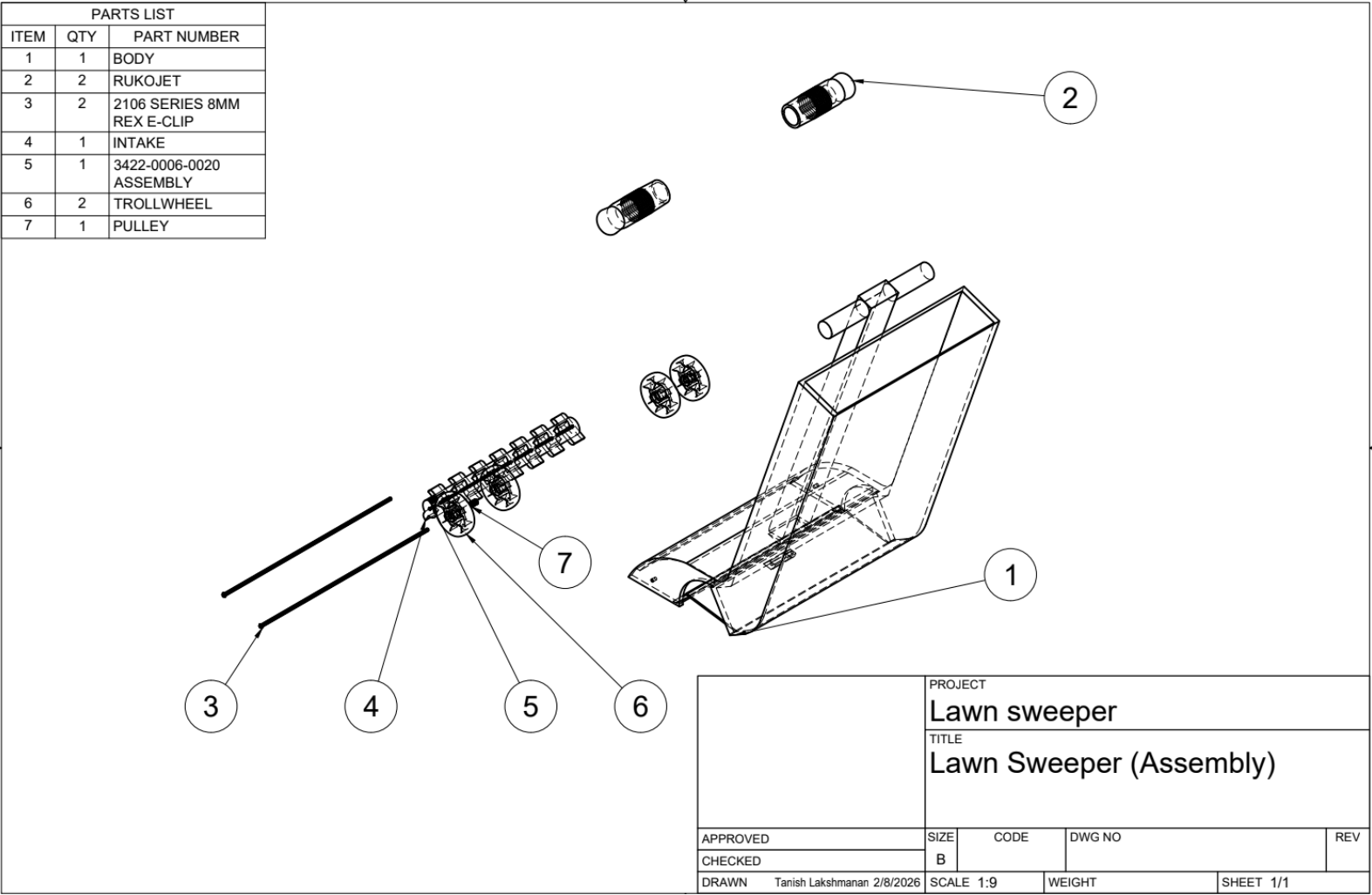


Figure G.1: Labeled exploded view

ITEM	QTY	PART NUMBER
1	1	BODY
2	2	RUKOJET
3	2	2106 SERIES 8MM REX E-CLIP
4	1	INTAKE
5	1	3422-0006-0020 ASSEMBLY
6	2	TROLLWHEEL
7	1	PULLEY

Figure G.2: Parts List

Tools and Equipment

The tools and equipment necessary to assemble the lawn sweeper are summarized in the table below. These items provide everything needed to accurately, safely, and efficiently complete the assembly of the design.

Item	Use	Source
2.5 mm screwdriver	Tightening the screws used in assembly	Neuqua Valley
PLA Glue	Attaching the handle to the body	Neuqua Valley
3D Printer	Printing custom parts	Neuqua Valley
Rubber Mallet	Pressing bearings into holes	Neuqua Valley
Ruler & Dial Caliper	Measurement	Neuqua Valley
Wrench/ Nut Driver	Hold and tighten nuts while installing screws	Neuqua Valley
Clamps	Holding parts steady when assembling parts	Neuqua Valley
Sandpaper	Smoothing out 3D Printed parts or adjusting parts for a smoother fit	Neuqua Valley

Needed Knowledge

The final product for the user will be mostly assembled, needing minimal external knowledge to put it together. However, constructing a prototype requires the team to apply a set of fundamental hands-on skills. The knowledge areas listed in the table below are necessary to successfully build and assemble a prototype for the Lawn Sweeper.

Item	Need Outside Assistance (Y/N)	Source	Outside Source Notes
Mechanical Assembly	N	Tanish Shreyas Seth	Group members
Belt and Pulley Systems	N	Tanish Shreyas Seth	Group members
3D Design Modeling	N	Tanish	Group members
3D Printing Operation	N	Tanish	Group members

Build Procedure

The build procedure contains detailed, step-by-step instructions for assembling the lawn sweeper.

3D print all required parts (hood, body, frame, handle, wheels (2), intake wheel, and intake claws) and gather all hardware (bearings, 624 mm axle, pulleys, belt, shaft collars, screws, nuts, and tools)

Hood & Intake Assembly

1. Press a bearing into each first hole on the 3D-printed hood, with the flanged side facing outward on both sides, as shown in Figure G.3.

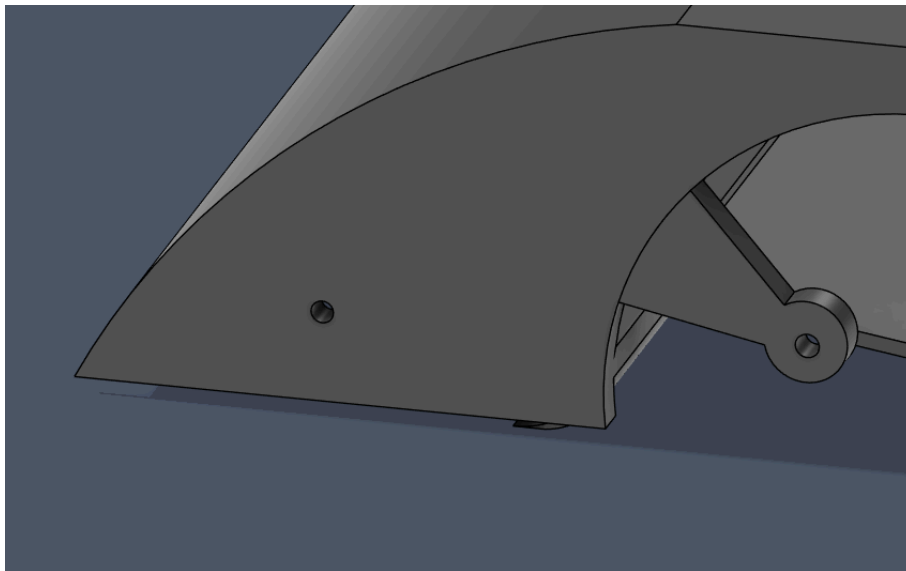


Figure G.3: Hood with bearing hole

2. Slide the 624 mm goBilda axle partially through the first bearing, as shown in Figure G.4.

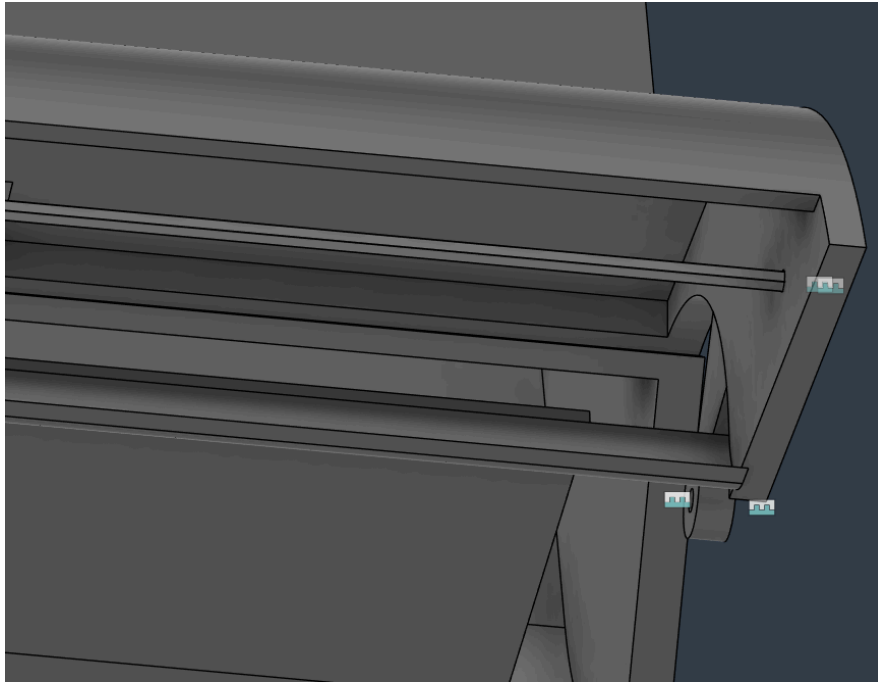


Figure G.4: Axle on the inside of the hood

3. Install the 8 mm 20-tooth pulley onto the axle near the inner wall, add the pulley and the belt, and position the pulley 0.45 inches from the inner wall with the set screw facing the wall and securely tighten, as shown in Figure G.5.

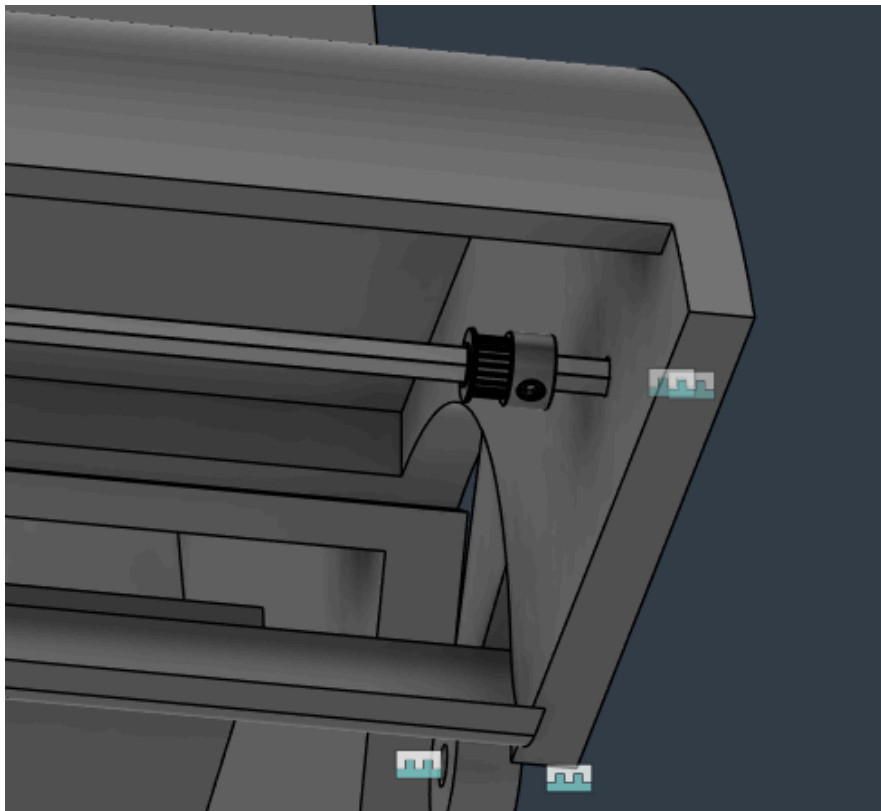


Figure G.5: 20-tooth pulley and belt attached

Intake Wheel Assembly

4. Attach each 3D printed intake claw to the 3D printed intake claw as shown in Figure G.6.

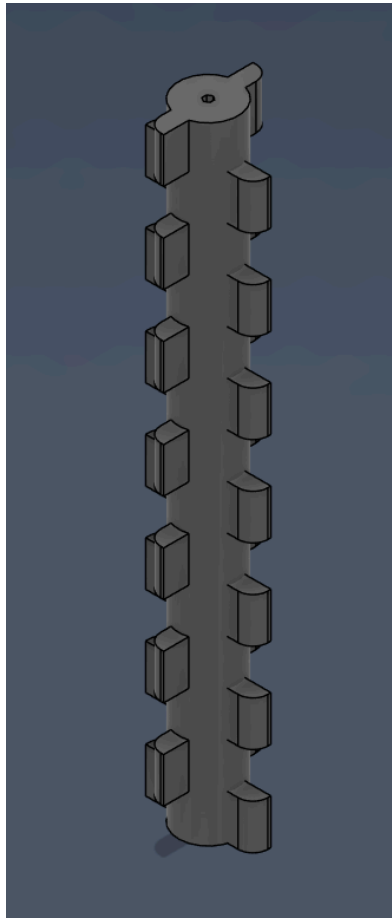


Figure G.6: Intake claws

5. Slide the intake wheel onto the axle, making sure the rounded edge of the claws is facing outward
6. Align the axle with the second bearing hole
7. Press bearings into the second set of holes in the hood, making sure the flanged side is facing outward.

Body Assembly

8. Slide the axle through a 3D printed wheel and add a shaft collar to secure, as shown in Figure G.7

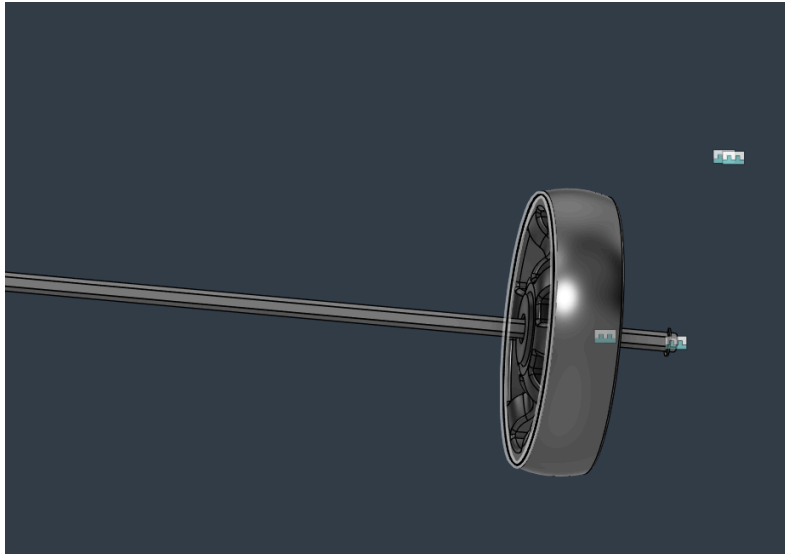


Figure G.7: Wheel attachment

9. Install the 40-tooth pulley onto the axle from the end and loop the belt around it as shown in Figure G.8

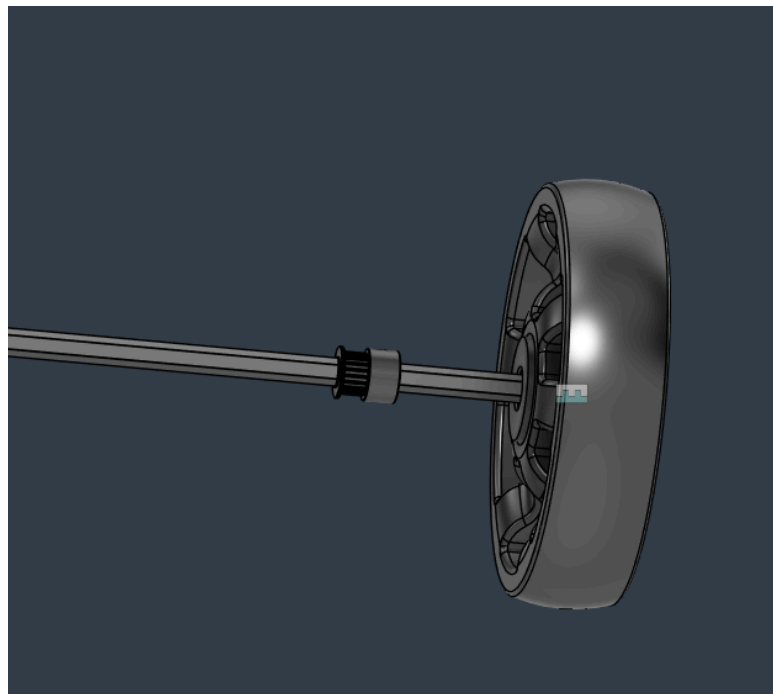


Figure G.8: 40-tooth Pulley installed

10. Slide the axle through the center holder of the 3D printed body as shown in Figure G.9.

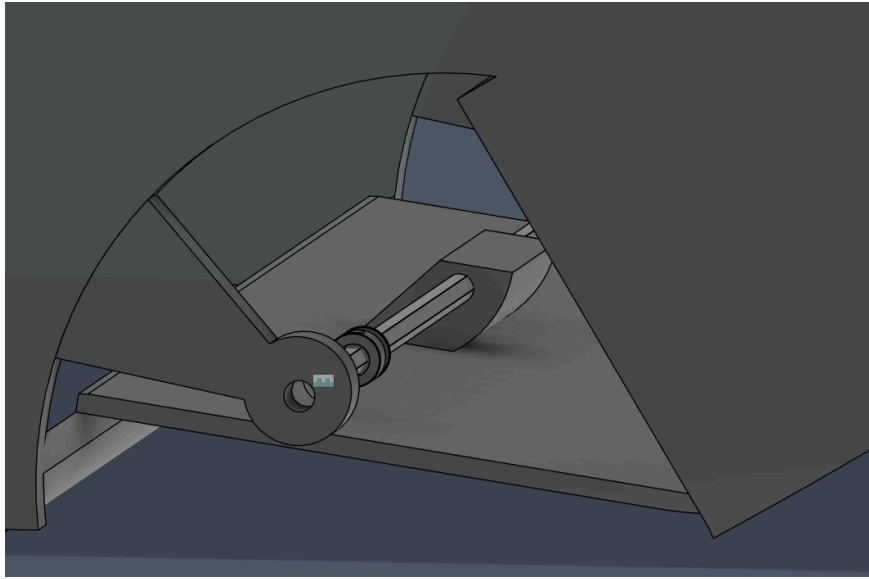


Figure G.9: Axle through the center holder

11. Add the second shaft collar and slide on the second 3D printed wheel

12. Push the axle through the final bearing and tighten all shaft collars and pulleys

Final Assembly

13. Attach the handle to the body using PLA glue as shown in Figure G.10.

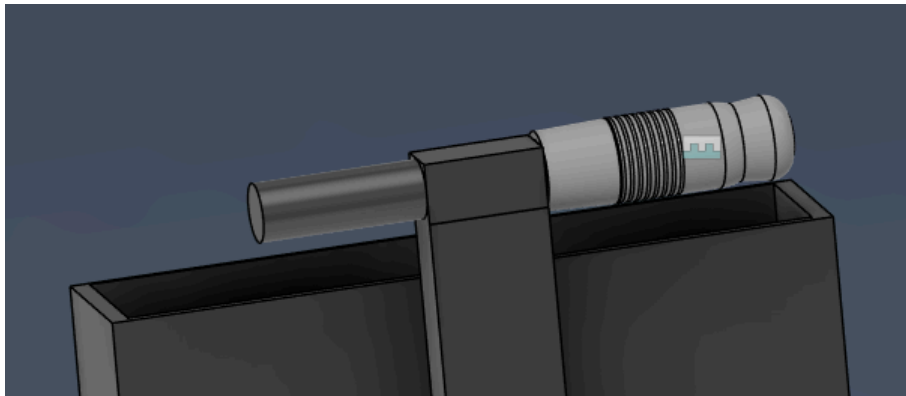


Figure G.10: Handle attached to body

14. Secure the body to the hood using four 8 mm screws and nuts with a 2.5 mm screwdriver

15. Attach the 3D-printed frame to the body using eight 8 mm screws with a 2.5 mm screwdriver

16. Drape the leaf bag over the 3D-printed frame. The final product should now look like Figure G.11.

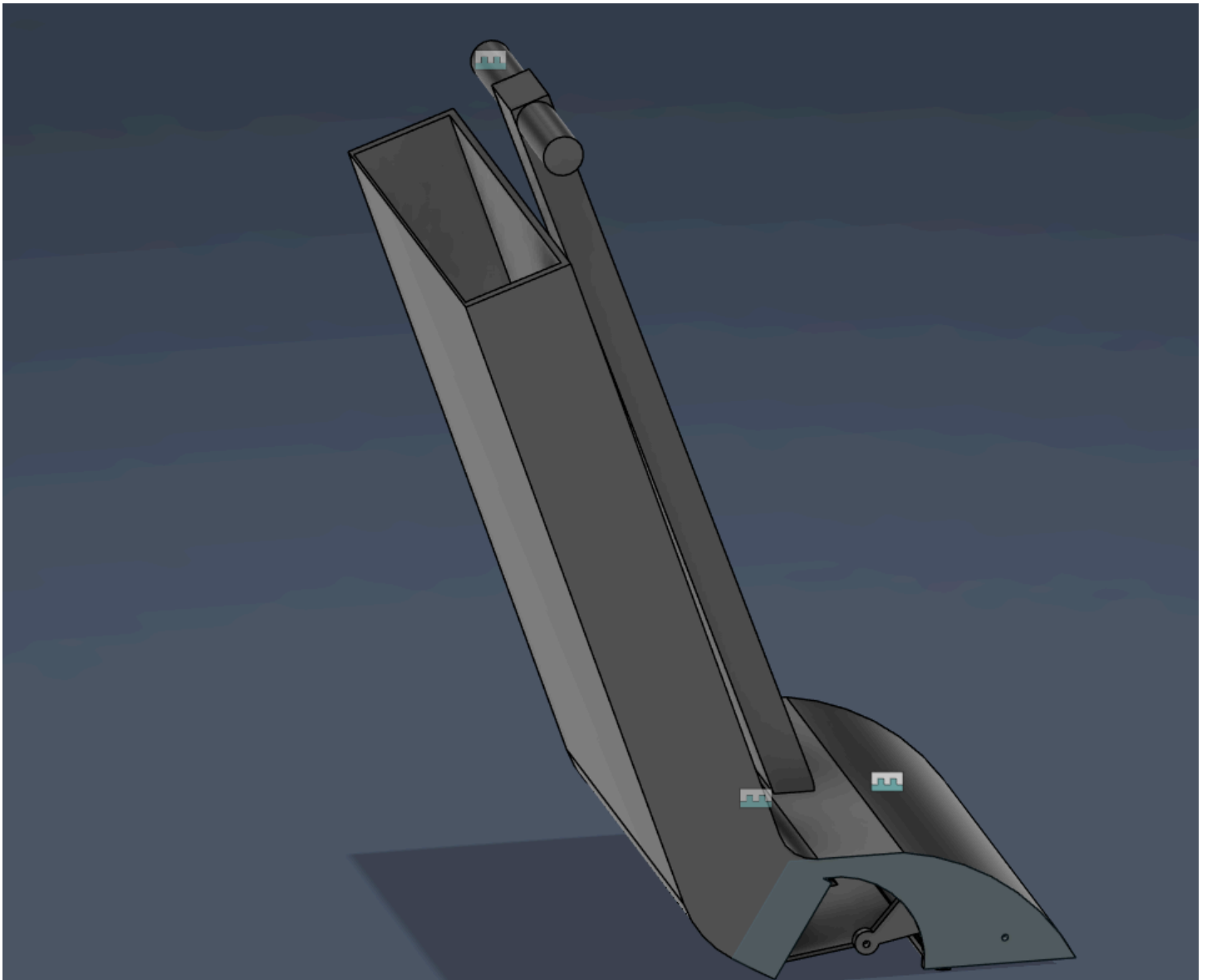


Figure G.11: Final Product